

# Monika Battula (Software Engineer)

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## WORK EXPERIENCE

### Machine Learning & Full Stack Engineer - Intern

Vertex AI

Jan 2025 – March 2025

San Francisco, USA

- Vertex labs, an AI-integrated platform that connects startups with investors through LinkedIn, Twitter, etc.
- Working on integration of email pipeline and Email automation.
- Tech Stack: React js, MongoDB, Vite js, API's, Web Scraping( Python, Selenium, Beautiful Soup), RAG

### Machine Learning & Drone - Intern

NIT Calicut

May 2024 – July 2024

Kozhikode, India

- Designed advanced pest management algorithms, reducing pesticide use by 25
- Implemented real-time object detection & precision cutting, improving productivity by 30
- Tech Stack: Raspberry Pi, Matplotlib, YOLO, DBSCAN, GMM, Simulink, Gazebo(RViz), RL, Pandas.

### Software Engineer Virtual Experience - Network Security

Goldman Sacs

May 2023 – July 2023

Bangalore, India

- Conducted IT security assessments and proposed upgrades to password protocols, reducing vulnerabilities by 60
- Developed comprehensive IT security assessments leading to the identification of three critical areas for improvement in password management; recommendations directly resulted in a measurable reduction of account compromise incidents

## OPEN SOURCE PROJECTS

### Software For Video Dubbing (Python, Flask, Whisper, IndicTrans2, gTTs)

- Built a web application for dubbing videos from English to various Indian regional Languages.
- Automated speech-to-text and text-to-speech processing using Whisper and IndicTrans2.
- Integrated speaker gender and tone detection for enhanced voice synthesis and accuracy.

### Email Automation-using-RAG (Python, AI Agents, RAG, Gmail API, Groq API, Google Gemini API)

- Automated Gmail inbox monitoring with AI agents to classify emails (complaints, inquiries, feedback, unrelated) and filter irrelevant messages.
- Generated personalized responses using LangGraph and RAG for accurate, context-aware handling of customer queries.
- Built an AI-driven QA pipeline to validate response quality, formatting, and relevance before client delivery.

## TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C++, C, JavaScript,
- **Frameworks & Tools:** SQL,MongoDB,React.js, Node.js, LangChain
- **Machine Learning:** TensorFlow, PyTorch, Hugging Face, LoRa , Scikit Learn, Automation, NLP, RL
- **Cloud and DevOps:** AWS, GCP, Azure, Docker, Kubernetes, GitHub Actions, CI/CD
- **Software Engineering:** OOP, Design Patterns,Agile, Quantum Computation, Service Oriented Computing

## EDUCATION

### University of Maryland Baltimore County

Master of Science in Computer Science

August 2025 – May 2027

### Gayatri Vidya Parishad College of Engineering

Bachelor of Engineering in Information Technology (Honors in IT)

August 2021 – March 2025

## PROJECTS

### Smart DDoS Defender( Python, Fast API, Graphana, Random Forest, Nginx)

- Engineered a Cloud-based solution for real-time DDoS detection & mitigation.
- Tech Stack: TensorFlow, NumPy, Pandas, Docker, Hping3, Pyflowmeter, Linux.

### DNS Packet Capture and Fields Parsing (Byte Level Malicious Detection)

- Engineered a sophisticated packet sniffer using raw sockets and pcap API to intercept Ethernet frames and analyze IP packets impacting 200+ network devices.
- Tech Stack: TensorFlow, NumPy ,C, raw socket programming, Berkeley Packet Filter (BPF).

## CERTIFICATIONS

- AWS Certified Cloud practitioner.
- Earned Certifications in CCNA, JNCIA-Junos, Fortinet.

Feb 2025

Jan 2024- Dec 2024

## PUBLICATION

- Published research paper on DDoS security Mechanism in IEEE.
- Published disease detection in apples using ML in IEEE.

March 2025

Oct 2024